

• Solve $\boxed{\frac{dy}{dx} + y = e^x} \rightarrow$ linear diff Eqn

Hint:- $dy + (y - e^x) dx = 0$

Hint (2) :- Is it Exact? NO

$$M = y - e^x \quad N = 1$$

$$\frac{\partial M}{\partial y} = 1 \quad \frac{\partial N}{\partial x} = 0$$

$$\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{1(1-0)}{1} = 1$$

$$I.f. = e^{\int 1 \cdot dx} = e^x$$

Exact diff Eqn

$$e^x(y - e^x) dx + e^x dy = 0$$

Solution

$$\int e^x y - e^x \cdot e^x dx = C$$

y constant

$$y e^x - \int e^x \cdot e^x dx = C$$

$$y e^x - \frac{(e^x)^2}{2} = C$$

$$e^x = z$$

$$e^x dx = dz$$

$$\int z dz = \frac{z^2}{2}$$

$$= \frac{(e^x)^2}{2}$$

• $\frac{dy}{dx} + p(x)y = r(x) \rightarrow$ linear diff Eqn.

$$I.f. = e^{\int p(x) dx}$$

solution of linear diff Eqn

$$\int r(x) I.f. dx + C$$

solution of linear with -

$$p(x) \cdot I.F = \int r(x) \cdot (I.F) dx + C$$

→ This diff Egn is linear in dependent variable y and $\frac{dy}{dx}$.

• $p(x)$ and $r(x)$ may contain higher powers of x .

* linear diff Egn is called Homogeneous if $r(x) = 0$.
Homogeneous linear diff Egn has form.

$$\frac{dy}{dx} + p(x)y = 0$$

* linear diff Egn is called Non-Homogeneous if $r(x) \neq 0$.

• How to solve Homogeneous linear diff Egn

$$\frac{dy}{dx} + p(x)y = 0$$

$$\frac{dy}{y} = -p(x) dx$$

$$\int \frac{dy}{y} = \int -p(x) dx + C$$

$$\ln|y| = \int -p(x) dx + C$$

$$y = C' e^{\int -p(x) dx}$$

Example:-

$$(1+x^2) \frac{dy}{dx} + y = 0$$

Example:-

$$(1+x^2) \frac{dy}{dx} + y = 0$$

$$\frac{dy}{dx} + \frac{1}{(1+x^2)} y = 0$$

$$\int \frac{dy}{y} = \int \frac{-1}{1+x^2} dx + C$$

$$\ln|y| = -\tan^{-1}x + C$$

$$y = C' e^{-\tan^{-1}x}$$

* How to solve non-homogeneous linear diff eqn.

$$\frac{dy}{dx} + p(x)y = r(x)$$

$$dy + (p(x)y - r(x)) dx = 0$$

$$M = p(x)y - r(x)$$

$$N = 1$$

$$\frac{\partial M}{\partial y} = p(x)$$

$$\frac{\partial N}{\partial x} = 0$$

If $p(x) \neq 0$, Not Exact.

$$\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{1}{1} (p(x) - 0) = p(x)$$

$$\underline{\underline{\text{I.f.}}} \quad e^{\int p(x) dx}$$

$$e^{\int p(x) dx} (p(x)y - r(x)) dx + e^{\int p(x) dx} dy = 0$$

Is it exact!?

$$M = e^{\int p(x) dx} (p(x)y - r(x))$$

$$\frac{dM}{dy} = e^{\int p(x) dx} p(x)$$

$$N = e^{\int p(x) dx}$$

$$\frac{dN}{dx} = e^{\int p(x) dx} \cdot p(x)$$

Exact diff solution

Solution is $\int e^{\int p(x) dx} p(x) y dx - \int r(x) e^{\int p(x) dx} dx = C$
 y constant

$$y \int e^{\int p(x) dx} p(x) dx = \int r(x) e^{\int p(x) dx} dx + C$$

$$z = \int p(x) dx$$

$$dz = p(x) dx$$

$$\int e^z dz = e^z$$

$$y \times e^{\int p(x) dx} = \int r(x) e^{\int p(x) dx} dx + C$$

$$y \times \text{I.f} = \int r(x) \times \text{I.f.} dx + C$$

• $\frac{dy}{dx} + p(x)y = r(x)$ is linear diff eqn.
 $\text{I.f} = e^{\int p(x) dx}$
Solution is
 $y \times \text{I.f} = \int (r(x) \text{I.f}) dx + C$

Question:- $\frac{dy}{dx} = \frac{1}{1+x^3} - \left(\frac{3x^2}{1+x^3} \right) y$

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$$\frac{dy}{dx} + \left(\frac{3x^2}{1+x^3} \right) y = \frac{1}{1+x^3}$$

$$p(x) = \frac{3x^2}{1+x^3}$$

$$r(x) = \frac{1}{1+x^3}$$

$$\begin{aligned} \text{I.f} &= e^{\int \frac{3x^2}{1+x^3} dx} \\ &= e^{\ln|1+x^3|} = 1+x^3 \end{aligned}$$

$$1+x^3 = z$$

$$3x^2 dx = dz$$

$$\int \frac{1}{z} dz = \ln|z|$$

Solution

$$\begin{aligned} y \times (1+x^3) &= \int \frac{1}{1+x^3} \cdot (1+x^3) dx + C \\ &\Rightarrow (1+x^3)y = x + C \end{aligned}$$

• $\frac{dy}{dx} + 2y = \sin x$

Solution:-

$$p(x) = 2$$

$$r(x) = \sin x$$

$$\text{I.f} = e^{\int 2 dx} = e^{2x}$$

Solution is

$$y e^{2x} = \int \sin x e^{2x} dx + C$$

(please complete solution)

• A curve is passing through the origin and at a point $R(x, y)$, where

• A curve is passing through the origin at a point $R(x, y)$, where slope of the tangent is given as $\frac{x^4 + 2xy + 1}{1 - x^2}$ for $-1 < x < 1$. What will be the Equation of the curve ??

$$\frac{dy}{dx} = \frac{x^4 + 2xy + 1}{1 - x^2}$$

$$y(0) = 0$$

When $x=0$,
 $\Rightarrow y=0$

$$\frac{dy}{dx} - \frac{2x}{1 - x^2} y = \frac{x^4 + 1}{1 - x^2}$$

$$p(x) = \frac{-2x}{1 - x^2}$$

$$q(x) = \frac{x^4 + 1}{1 - x^2}$$

I.F. $e^{\int p(x) dx} = e^{\int \frac{-2x}{1 - x^2} dx} = 1 - x^2$

Solution

$$y(1 - x^2) = \int \frac{x^4 + 1}{1 - x^2} x(1 - x^2) dx + C$$

$$y(1 - x^2) = \frac{x^5}{5} + x + C \quad (\text{complete solution})$$

$x=0, y=0$

$0 = C \Rightarrow$ particular solution
 $y(1 - x^2) = \frac{x^5}{5} + x$

Question :- Solve $(x + 2y^3) \frac{dy}{dx} = y$

$$\frac{dy}{dx} = \frac{y}{x+2y^3}$$

$$\Rightarrow \frac{dx}{dy} = \frac{x+2y^3}{y}$$

$$\Rightarrow \frac{dx}{dy} - \frac{1}{y} x = 2y^2$$

$$p(y) = -\frac{1}{y} \quad r(y) = 2y^2$$

$$\text{I.f.} \quad e^{\int -\frac{1}{y} dy} = e^{-\log y} = \frac{1}{y}$$

Solution is

$$x \cdot \frac{1}{y} = \int 2y^2 \cdot \frac{1}{y} dy + C$$

$$x/y = y^2 + C$$

Question:-

Solve

$$x \frac{dy}{dx} = y-1,$$

$y(0) = 1$ → initial condition
 $x=0, y=1$

Initial value problem.

→ infinitely many solutions

$$\int \frac{dy}{y-1} = \int \frac{dx}{x} + C$$

$$\ln|y-1| = \ln|x| + C$$

$$y-1 = Cx$$

$$x=0, y=1 \rightarrow \underline{Cx + y = 1}$$

New Initial condition $y(0)=0 \rightarrow x=0, y=0$
No solution $\leftarrow -1=0$ Not true